Mock Interview Edition

Visual Search: One Embedding for Three Products

17 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

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07 Float, low-dimensional float, or binary? Design choice 12
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13 How many replicas at peak? Capacity 21
17 Ten times the images and the traffic Scale 27

SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: unification, evidence and launch decisions

Deciding whether one embedding should replace three, reading the evidence for it honestly, and choosing how to launch.

01 Three embeddings, or one?

FRAMING

SCENARIO

A company runs three visual search products. Each has its own embedding model, training pipeline and backfill, over about 5 billion images. Leadership proposes replacing them with one shared embedding. The Lens team objects that their metric will suffer.

OPENING PROBE

What does each design cost?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the paper say was the pain?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What would you need to see before agreeing with unification?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does unification not remove?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 Was it multi-task training, or just a better backbone?

EVIDENCE

SCENARIO

The paper's Table 4 shows the unified embedding beating each old specialized embedding: Shop-the-Look P@1 33.0 to 52.8, Flashlight avg P@20 53.4 to 60.2, Lens avg P@20 17.8 to 18.4. A colleague says: “That proves training one model on all three datasets is better than training three.”

OPENING PROBE

What does Table 4 compare?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why can't multi-task training take the credit?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Which table isolates multi-task training?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What else would you want before trusting the +3.6?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 Offline says +0.6, human raters say +110%

EVALUATION

SCENARIO

For Lens, the paper's offline evaluation moves from avg P@20 of 17.8 to 18.4. Human raters comparing the same two systems report a P@5 change of +110.1%, and the online test shows clickthroughs up 35%. A reviewer asks which number is right.

OPENING PROBE

How big is each change?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why might the offline number understate the gain?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the human judgment add?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Which do you use for what?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 Launch Shop-the-Look without an A/B test?

LAUNCH DECISION

SCENARIO

Shop-the-Look had not launched to any user when the unified embedding was ready, so no A/B test could be run. The team has human judgments. A director asks whether that is enough to ship.

OPENING PROBE

What evidence did the team have?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What can raters not tell you?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Is it enough to ship?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What do you do afterward?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

The model: shared backbone, classification heads and binary embeddings

How a classifier produces a similarity space, how the four tasks share one trunk, and what binarizing costs.

05 Can a classifier train an embedding?

MECHANISM

SCENARIO

A teammate says: “Embeddings need a triplet or contrastive loss. A softmax classifier cannot give you a similarity space.” The paper trains its embedding with four classification heads.

OPENING PROBE

What is a proxy, and what do the heads do?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why does that produce a useful similarity space?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why not triplets?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How does it cope with 15,000 or 50,000 classes?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

06 A noisy head in a shared trunk

DEBUGGING

SCENARIO

A taxonomy change leaves 8% of Shop-the-Look product-class labels wrong (illustrative). In the next retrain, Flashlight's avg P@20 falls by 1.5 points, even though Flashlight's data, head and evaluation set did not change.

OPENING PROBE

Why would Flashlight move?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How do you confirm it is the cause?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What do you do about it?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How would you have caught it earlier?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

07 Float, low-dimensional float, or binary?

DESIGN CHOICE

SCENARIO

To cut storage, someone proposes keeping float32 embeddings but shrinking them to 64 dimensions. Another proposes thresholding the 2,048-dimensional embedding to bits. A third says binarizing always costs accuracy, so keep floats.

OPENING PROBE

How do the options compare in storage?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What did the paper's earlier work find?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Does binarizing cost accuracy?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What fixed it?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

How do you decide on your data?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

08 Equal loss weights and uniform dataset mixing

TRAINING DESIGN

SCENARIO

The three datasets have very different sizes: about 800K, 540K and 340K images. The paper mixes them equally in every batch and gives the four losses equal weights. A reviewer says: “Weighting by dataset size or learning the weights would obviously be better.”

OPENING PROBE

What did the paper do, and try?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does uniform mixing mean for the small datasets?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why might equal loss weights be wrong in general?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How would you test the reviewer's claim on your data?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

PART 3

The index and the query path: storage, shards and search

Sizing a binary index, checking a scan against the hardware, and what scatter-gather does to latency and to resharding.

09 Size the index

SIZING

SCENARIO

A storage engineer asks how much space the visual search index needs, and whether the shift from the old embeddings to the unified binary one changes the plan. Assume 5 billion images and a 2,048-dimension unified embedding (illustrative).

OPENING PROBE

How big is the index in each format?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How many shards?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the 32 times depend on?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the plan leave out?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 14

10 Review the per-shard scan

CODE REVIEW

SCENARIO

An engineer ships the function below to search one shard of 312.5 million 256-byte codes. In staging, the process runs out of memory, and when it does not, a query takes tens of seconds. The code gives correct results on a test of 1,000 rows.

```
import numpy as np

def shard_top_k(query_bits, corpus_bits, k=20):
    # query_bits: (256,) uint8; corpus_bits: (N, 256) uint8
    xor = np.bitwise_xor(corpus_bits, query_bits)
    distances = np.unpackbits(xor, axis=1).sum(axis=1)
    return np.argsort(distances)[:k]
```

OPENING PROBE

What is wrong with the memory use?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What is wrong with the time?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

Does the fix make exact scanning viable at 926 queries a second?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What would you test?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

11 Add a 17th shard

RESHARDING

SCENARIO

The index has outgrown 16 shards. The plan is to add a 17th. Rows are placed by image_id modulo the number of shards. An engineer says the move is cheap because only one shard is new.

OPENING PROBE

What moves under modulo placement?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What would consistent hashing move?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does a query see during the move?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Does placement matter for query correctness?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

12 Fan-out makes the tail worse

SCATTER-GATHER

SCENARIO

Each shard answers within 100 ms 99% of the time. A query fans out to all 16 shards and waits for every one. Users complain that search is often slower than 100 ms, though no shard's dashboard shows a problem.

OPENING PROBE

What fraction of queries are slow?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How does it grow?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What can you do about it?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you monitor?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

13 How many replicas at peak?

CAPACITY

SCENARIO

The index is 16 shards chosen by capacity. Peak traffic is 926 searches a second. A colleague says 16 machines are enough because each holds 80 GB of codes. Another proposes brute-force scanning, because binary codes are cheap to compare.

OPENING PROBE

What load does each shard see?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the brute-force scan need?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does an index change?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What must you measure for the index?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

PART 4

Evaluation, operations, risk and scale

Shipping a new embedding safely, building evaluation sets you can trust, what to protect, and what ten times the size breaks.

14 A retrain ships and binary quality drops

OPERATIONS

SCENARIO

A new training run improves the float embedding on every product's evaluation set. The team is ready to start a full backfill of the binary index. A reviewer asks what has been checked on the binary version.

OPENING PROBE

What should be checked before the backfill?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How do you ship it in steps?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does redoing a backfill cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What gates the rollout?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

15 Can you trust the evaluation set?

EVALUATION

SCENARIO

A new embedding raises Shop-the-Look P@1 from 49.2 to 52.8 on the paper's evaluation set of 600 query objects searched in a corpus of 51,421 images. The team wants to ship on that result.

OPENING PROBE

What are the sets?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How noisy is a P@1 on 600 queries?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does a small corpus hide?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How do you keep the sets useful?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 20

16 Camera photos as queries: privacy and abuse

RISK

SCENARIO

Lens lets users point a phone at anything and search with the photo. A security reviewer asks what the system stores, who can see it, and how it could be abused. The paper does not discuss these.

OPENING PROBE

What is sensitive here?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What should the policy be?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How could the index be abused?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What defenses help?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 21

17 Ten times the images and the traffic

SCALE

SCENARIO

Imagine the corpus grows to 50 billion images and peak traffic to 9,259 searches a second. A manager asks what breaks first, and whether to add machines or change the design.

OPENING PROBE

What does ten times do to storage and shards?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does it do to the query path?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does it do to the backfill?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Which lever for the fan-out?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	Three embeddings, or one?	5			
02	Was it multi-task training, or just a better backb...	5			
03	Offline says +0.6, human raters say +110%	5			
04	Launch Shop-the-Look without an A/B test?	5			
05	Can a classifier train an embedding?	5			
06	A noisy head in a shared trunk	5			
07	Float, low-dimensional float, or binary?	6			
08	Equal loss weights and uniform dataset mixing	5			
09	Size the index	5			
10	Review the per-shard scan	6			
11	Add a 17th shard	5			
12	Fan-out makes the tail worse	5			
13	How many replicas at peak?	5			
14	A retrain ships and binary quality drops	5			
15	Can you trust the evaluation set?	5			
16	Camera photos as queries: privacy and abuse	5			
17	Ten times the images and the traffic	5			